

Project Report

Uma Oya

Multi-purpose Development Project



**Project report by,
Group 1 students**

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01. Current Market Conditions

The Uma Oya Multipurpose Development Project significantly impacts Sri Lanka's energy, water, and local economic sectors.

Energy Sector:	• Adds approximately 120 MW hydro capacity (≈230 GWh/year expected) • Reduces dependency on costly thermal power and imported fuel. • Project delays caused financial losses due to missed generation capacity.
Agriculture:	• Transfers water from Uma Oya to Kirindi Oya basin, irrigating 4,500–6,000 ha
Local Economy and Transport:	• Created demand for construction materials, machinery, and skilled labor. • Caused temporary road closures and local disruptions due to tunneling.
Competition and Pricing:	• Hydropower indirectly competes with thermal power by reducing electricity costs. • Delays increase reliance on imported fuel, raising overall electricity costs.
Suppliers and Contractors:	• Project managed by the Ministry of Irrigation with technical support from CEB. • Major contractors include FARAB (Iran) and local subcontractors. • Financed by the Export Development Bank of Iran.

02. Stakeholders (Demand and Supply Sides)

Demand-Side Stakeholders: • Ceylon Electricity Board (CEB) • Farmers and Irrigation Beneficiaries • Households and Industries	Supply-Side Stakeholders: • Ministry of Irrigation • FARAB and subcontractors • Export Development Bank of Iran	Other Stakeholders: • Local Communities • Environmental Authorities and Civil Organizations • Judiciary and Regulators
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03. Areas for Policy Intervention

A. Strengthening Environmental and Social Governance:

Mandate independent EIA and SIA reviews before project expansions.
Ensure transparency in public disclosure of impact assessments.

B. Compensation and Livelihood Restoration:

Establish clear compensation criteria and rapid grievance redress mechanisms.

C. Technical Risk Mitigation:

Continuous geotechnical and groundwater monitoring to prevent disasters.

D. Integrated Basin Water Management:

Introduce water allocation regulations ensuring minimum environmental flows.

E. Financial and Market Reforms:

Conduct independent audits for cost overruns and establish contingency funding.

F. Energy Market Integration:

Align hydro dispatch and tariff structures to maximize renewable energy value.

04. Desired Outcomes

Primary Outcomes	• Reliable renewable energy generation (120 MW; 230 GWh/year). • Enhanced irrigation and potable water supply for southern dry regions. • Reduced national thermal fuel costs and carbon emissions.
Social and Environmental Outcomes	• Restored groundwater levels and fair compensation for affected households. • Long-term infrastructure safety and stability.
Measurable Indicators	• MW commissioned and GWh generated annually. • Hectares irrigated and households receiving water supply. • Reduction in thermal generation and fuel imports.

05. Goals & objectives

1. Provide reliable, low-cost renewable electricity

- Install underground power stations
- ensure dry-season availability via reservoir/headrace design and O&M
- minimize greenhouse emissions by reducing use of fossil fuels.

2. Expand irrigated area and improve output in dry zones

- Supply irrigation to ~6,000 ha; ensure seasonal water reliability
- Build supporting infrastructure like canals and rehabilitate tanks

3. Provide potable and industrial water, supporting socio-economic development

- Meet annual water demand; set affordable domestic tariffs and cost-recovery industrial tariffs

06. Project cost

1. Initial planning cost

In the Uma Oya Project, SLR 1.612 billion was allocated for planning, including engineering, design, feasibility studies, consultations, and construction supervision.

2. Construction cost

The total project cost is estimated to be approximately US\$529 million (approximately Rs. 80 billion)

- **Permanent Structures & Foundation:** Rs. 29.8 billion for reservoirs, tunnels, civil works, materials, transport, and labor.
- **Mechanical & Power Systems:** Rs. 21.7 billion for machinery, equipment, and electro-mechanical installations.

3. Operation and Management Cost

The Uma Oya Multipurpose Development Project is expected to cost Rs. 3.17 billion a year to operate and maintain. All ongoing expenses necessary to guarantee the uninterrupted, secure, and effective operation of the irrigation and hydropower systems are included in this allocation.

07. Sources of investment

- **Export Development Bank of Iran (EDBI)**
- **Srilankan government**

85% of the \$ 529 million project was initially to be funded by Iran. However, after 2013 economic sanctions were imposed on that country, Sri Lankan government undertook to fund the bulk of the project.

08. Financial sustainability

The Uma Oya Project achieves financial sustainability through revenue from hydropower, irrigation, and water supply, combined with cost efficiency and long-term economic benefits.

Once completed, the project is expected to save the CEB about Rs. 80 million per day, or roughly Rs. 1.5 billion per month, compared to costlier power sources..

Addition of 120 MW to national grid helps reduce dependence on fossil fuel power.

Power generation

The 120 MW power plant produces around 290 GWh of electricity annually, which is sold to the national grid. This steady revenue helps recover investment costs and cover operational and maintenance expenses.

Irrigation and water supply

supply of drinking and industrial water creates potential for direct revenue collection from water tariffs

Pricing structure

Pricing structure ensures steady revenue generation from power, water, and irrigation services, thereby supporting the long-term financial sustainability of the Uma Oya Project.

Electricity Tariff :

Power generated from the Uma Oya project is sold to the national grid at a cost-reflective tariff, ensuring sufficient revenue to cover operation, maintenance, and loan repayment.

Water Tariff (Domestic and Industrial Use) :

A step-by-step pricing system is used, domestic users pay lower prices, while industries pay higher prices to keep the service affordable and able to cover its costs.

09. Social and Environmental Impacts & the ways of prevent and Internalize

The project implementation has had catastrophic localized costs, positioning the UOMDP as a critical study of governance and geo-engineering failure in large-scale infrastructure development.

Environmental and Geo-Hydrological Impacts

A. The Hydrogeological Crisis of Deep Tunneling

The major environmental failure occurred during tunneling through the shallow aquifer in Bandarawela, causing a water inflow of 450 L/s and drying over 3,000 wells, streams, and wetlands. Despite sealing attempts, inflow persisted at 110–125 L/s, confirming continued aquifer collapse.

B. Ecosystem degradation and watershed vulnerability

Beyond hydrological issues, the UOMDP has caused chronic ecosystem damage in the Uma Oya watershed. Construction, roads, and resettlement led to habitat loss, fragmentation, and degradation in four protected areas, threatening species survival and biodiversity..

In addition, the conversion of forests and other vegetated lands to agricultural land and farmland has resulted in high levels of soil erosion, particularly on steep slopes. The reduction in vegetation cover and the expansion of cultivated land exacerbate soil loss, affecting water quality within the catchment area. Geo-fracturing is

also a significant concern. The changes to groundwater levels caused by the tunnels and the two proposed reservoirs pose a risk of creating conditions conducive to flooding and landslides.

The project's improved irrigation in Monaragala and Hambantota helps farming but also increases pesticide and fertilizer use. This can pollute rivers and food chains over time. So, while it solves the water problem upstream in Bandarawela, it creates new environmental and health issues downstream.

Social and Public Health Impacts

Displacement, Infrastructure Collapse, and Livelihood Destruction

Physical destruction has damaged over 7,000 homes along the tunnel route and displaced 1,500 families within their own homeland. The ongoing collapse caused by geotechnical instability has also destroyed essential social infrastructure, including schools and places of worship.

The project ruined the main economic base of upper catchment communities by destroying farmland and drying wells. Although resettlement plans gave temporary help like Rs. 15,000 rent and 500 liters of water weekly, these short-term measures couldn't replace the permanent loss of land and livelihoods, showing a major failure in rebuilding affected lives.

Severe Mental Health and Lowering of Public Trust on the Project

The water crisis led to serious mental health problems such as PTSD, depression, anxiety, and stress. Ongoing worries about money, housing, and water made things worse. It also reduced trust in health officials, making recovery and community strength harder.

Prevention Strategies: Technical and Ecological Risk Mitigation

Advanced Geo-Engineering Protocols and Adaptive Management

- Mandatory Pre-Excavation Grouting
- Integrated Monitoring and Transparency
- Route Redesign and Assessment

Comprehensive Ecological Restoration and Water Management

1. Stop severe degradation in the Uma Oya River Basin by restoring soil and water flow through reforestation with native Bandarawela trees.
2. Launch an artificial aquifer recharge program in Bandarawela to restore groundwater until natural recharge recovers.
3. Introduce strict IPM policies and organic farming subsidies in new irrigation areas to reduce downstream agrochemical pollution.

Internalization Mechanisms

1. Reforming the Compensation and Redress Framework

Establishment of an Independent Damage Assessment Board (IDAB)

Non-market damage assessment, Legal enforcement and remedies

2. Establishment of a permanent Project Impact and Remediation Fund (PIF)

3. Institutional and policy reforms for future infrastructure governance

Mandatory Independent Environmental Impact Assessment Review

Strengthening Regulatory Capacity and Integrity, Pre-emption project waiver clause

10. Economic Outcomes of the Intervention

This project brings significant financial benefits to the national economy with a net worth of Rs 24.7 billion.

Consumer Surplus (CS) after the Intervention

Consumer Surplus is the extra benefit from paying less than expected

- **Recreational Benefits (Uda Walawe National Park)**

In 2009, Uda Walawe National Park earned Rs. 6.64 million in recreation benefits. The Uma Oya Project is expected to raise visitors by 10%, adding Rs. 664,000 to the annual consumer surplus.

- **Increased Water Supply**

The Uma Oya Project will supply water to 25,000 acres of paddy fields, helping farmers grow more and spend less on water. These savings increase their consumer surplus.

- **Reduced Electricity Costs**

The project will add 120 MW of hydropower to the grid, providing cheaper, more reliable electricity and increasing consumer surplus.

- **Lower Food Prices**

Using Alikota Ara Tank water to irrigate old farms makes farming cheaper and boosts food supply, lowering prices and increasing consumer surplus.

Producer Surplus (PS) and Related Economic Benefits

Producer Surplus is the extra earnings from selling above the minimum price. In this project, it comes from hydropower and increased farming profits.

- **Benefits from Agricultural Expansion**

The project will irrigate 5,616 ha, allowing 1,500 ha of paddy land to grow in both seasons, boosting production, profits, and farmers' Producer Surplus.

- **Hydropower Generation**

Selling electricity generates revenue for the operator, increasing society's Producer Surplus.

- **Increased Fisheries**

The new Kuda Oya and Alikota Ara reservoirs add 497.4 ha of water, boosting fish growth, fishermen's income, and future fish stocking.

11. Identified Benefits and Annual Quantification

The project's first-year benefits include measurable gains in power, farming, and water supply.

Power and Energy Benefits

- **Installed Capacity:** The proposed Uma Oya power plant has an installed capacity of 120 MW.
- **Annual Energy Production:** The mean annual energy production is estimated to be 231 GWh.
- **Nature:** Hydropower is a low-cost, eco-friendly way to generate electricity.
- **Quantitative Value (Annual):** The benefit is valued using the "Avoided Cost Method" (savings from not building alternative thermal plants).
- **Avoided Thermal Power Generation (Economic Value):** Rs 20,710,488,000

Agriculture Benefits (Irrigation)

- **Water Allocation:**

The project moves 145 million m³ of water each year from Uma Oya to Kirindi Oya.

- **Expansion of Cultivation:**
Existing Lands: 1,500 ha of paddy will grow in both seasons, doubling intensity.
New Lands: 4,500–9,000 ha in Kirindi Oya basin gain irrigation for farming.
- **Crop Profitability: Water availability enables increased cultivation of Other Field Crops (OFC).**
Watermelon, capsicum, and green gram mix yield Rs 812,800/ha/year on 25% of land, while watermelon and capsicum yield Rs 730,000/ha/year on the rest.
- **Quantitative Value (Annual):**
The economic value from growing more paddy in the first year is Rs 84,780,000.

Water Supply Benefits

- **Water Volume:** The project supplies 30 million m³ of water annually for drinking and industrial use.
- **Quantitative Value (Annual):** The expected benefit is Rs 15 million at Rs 5 per 1,000 liters.
- **Other Economic and Social Benefits**

Benefit Area	Description	Quantitative Value (Annual Economic Value, First Year)
Increased Fisheries	Economic value is expected to double with year-round harvesting, especially in tanks like Handapanagala .	Rs 3,600,000
Increased Water Spreading Area	The project adds 497.4 ha of water from new reservoirs like Kuda Oya and Alikota Ara .	Rs 259,580,02

12. Evaluation of Economic BCA

Duration: 2011-2024 | **Total Investment:** USD 514 M | **Economic Life:** 30 years | **Discount Rate:** 10%

1. Cost analysis

Cost Category	Amount (USD million)
Capital expenditure	514
Operation and maintenance (Hydropower)	2.5 per year
Operation and maintenance (Irrigation)	1.8 per year
Total annual recurrent	4.3 per year
Economic Cost (Shadow Priced)	520

2. Benefit analysis

Benefit Category	Annual Value (USD million)	Description
Hydropower generation	34.8	290 GWh/year at USD 0.12/kWh
Irrigation Benefits	16.8	6,000 hectares at USD 2,800/ha crop value increase
Water Supply	5.2	Drinking & industrial water (Badulla, Monaragala, Hambantota)
Regional Development	8.5	Employment creation, trade enhancement
Environmental Benefits	3.2	Reduced emissions vs. thermal power generation
Food security	4.5	Import substitution, agricultural productivity

Total annual benefits = 73 USD million

Net Present Value

Present Value of Benefits: USD 821.4 million
Present Value of Costs: USD 568.7 million
Net Present Value (NPV): USD 24.7 billion

- $NPV = \sum [(Benefits - Costs) / (1 + r)^t]$
Where: $r = 10\%$ discount rate, $t = \text{year (0 to 30)}$

Economic Internal Rate of Return (EIRR)

- The EIRR is the discount rate at which $NPV = 0$, Through iterative calculation: **EIRR = 13.9%**

Benefit–Cost Ratio (BCR)

- $BCR = \text{Present Value of Benefits} / \text{Present Value of Costs}$
BCR = USD 821.4 million / USD 568.7 million = 1.44

Payback Period (PBP)

- With positive NPV and EIRR, the project is expected to recover costs in the medium to long term.

13. Financial Evaluation

An Extended CBA, including environmental values, assessed the Uma Oya Project using a 10% discount rate over a 50-year economic life to reflect Sri Lanka's capital costs.

Net Present Value (NPV)

Discount Rate	Net Present Value (Rs)
5%	120,273,809,536
10%	24,708,557,253
12%	9,362,977,518

- The project's economic viability and contribution to the national economy are confirmed by its positive net present value (NPV) of Rs 24.7 billion at a 10% discount rate.

Economic Internal Rate of Return (EIRR)

- The project's EIRR is 13.87%, which exceeds the 10% benchmark, making it a valuable investment.

Benefit–Cost Ratio (BCR)

- At a 10% discount rate, the Benefit–Cost Ratio (BCR) is 1.42. The project is financially viable, and yields benefits that exceed its expenses because this ratio is higher than 1.0.

Sensitivity Analysis

Condition	NPV (Rs, at 10%)	EIRR	BCR
Base case	24,453,868,267	13.84%	1.42
One-year delay in benefits	16,770,453,703	12.47%	1.29
10% cost overrun	19,454,132,110	12.84%	1.31
Combination (delay + overrun)	11,770,717,546	11.62%	1.19
10% increase in oil prices	30,326,283,983	14.65%	1.52

The project remains feasible under delays or cost overruns, with a resilient EIRR of 11.62%, above the 10% threshold.

Payback Period (PBP)

- There was no information regarding the project's Payback Period (PBP) in the sources.

14. Multi-Criteria Analysis (MCA)

MCA is a decision-making tool used to compare different options (projects, policies, investments) when there are many factors to consider, not just money.

Alternative Projects:

Small reservoirs and canals were built to improve local agriculture in Monaragala and Badulla districts. Cost-effective, low environmental impact, but no energy production.

Local Irrigation Expansion:

Small reservoirs and canals were built to improve local agriculture in Monaragala and Badulla districts. Quick, cost-effective, low environmental impact, but no energy production.

Solar + Small Dams:

Install solar farms and small dams for renewable energy and irrigation. Sustainable and modular, but high land use and limited water/energy benefits compared to UOMDP.

Improved Water Management:

Upgrade reservoirs, rainwater harvesting, and efficient irrigation systems. Minimal environmental and social risks, cost-effective, but energy benefits are very low.

Criteria

MCA is used to evaluate projects based on multiple economic, social, environmental, technical, and resource-related factors rather than cost alone.

Criterion	Description
Economic	Project's financial benefits and cost-effectiveness for stakeholders.
Environmental	Impact on natural ecosystems, water, soil, and biodiversity.
Social	Effects on local communities, livelihoods, and resettlement needs.
Technical	Feasibility, reliability, and engineering complexity of the project.
Time	Duration to complete and operationalize the project effectively.
Water Reliability	Ability to provide consistent, adequate water supply for use.
Energy	Electricity generation capacity and contribution to national grid.

The weights are given according to the importance of each criterion

Criteria	Economic	Environmental	Social	Technical	Time	Water Reliability	Energy	Total
Weight (%)	20	18	16	15	8	13	10	100

Weighted Total Score = $\sum (\text{Score} \times \text{Weight})/100$

Raw Scores (1–9 scale, 1 = poor, 9 = excellent)

Criteria	UOMDP	Local Irrigation	Solar + Small Dams	Water Management
Economic	8	6	5	4
Environmental	6	7	8	9
social	6	6	7	5
technical	7	7	6	8
time	5	7	6	
Water reliability	8	6	6	9
energy	9	3	6	4

MCA Comparison Matrix – UOMDP vs Alternatives

Criteria (weight/100)	UOMDP	Local Irrigation	Solar + Small Dams	Water Management
Economic (0.2)	8x0.2=1.6	6x0.2=1.2	5x0.2=1.00	4x0.2=0.8
Environmental (0.18)	6x0.18=1.08	7x0.18=1.26	8x0.18=1.44	9x0.18=1.62
Social (0.16)	6x0.16=0.96	6x0.16=0.96	7x0.16=1.12	5x0.16=0.8
Technical (0.15)	7x0.15=1.05	7x0.15=1.05	6x0.15=0.90	8x0.15=1.2
Time (0.08)	5x0.08=0.40	7x0.08=0.56	6x0.08=0.48	9x0.08=0.72
Water reliability (0.13)	8x0.13=1.04	6x0.13=0.78	6x0.13=0.78	4x0.13=0.52
Energy (0.10)	9x0.1=0.9	3x0.10=0.30	6x0.10=0.60	1x0.10=0.10
Total score	7.03	6.11	6.32	5.86

MCA compares UOMDP with three alternatives across seven criteria, with UOMDP scoring highest (7.03/9) for strong water and energy benefits.

15. Risk Assessment

When assessing acceptable risks, two main factors can be considered.

Seasonal water inflow variations – Minor inflow fluctuations from seasonal rainfall can be managed through reservoir operations and hydropower scheduling.

Moderate cost variations – Minor fluctuations in construction or material expenses are tolerated acceptable since there is contingency fund and control mechanisms of costs to ensure financial stability.

	Sensitivity Analysis	Risk Mitigation Strategies
Energy price fluctuations	Energy price changes affect the revenue and financial health of a 120 MW hydropower plant.	Sign long-term PPAs with the Ceylon Electricity Board, balance renewable energy sources, and use flexible tariffs to boost profit.
Price of agricultural products	Analyze how changes in crop prices and production levels affect irrigation revenue in Monaragala, Hambantota, and nearby areas.	Diversify the crops, introduce value added agro products, empower the farmer cooperatives and enhance market connections to minimize dependence on single crop.
Construction delays and cost overrun	Evaluate how higher material, labor, or equipment costs, and delays in tunneling or reservoir construction, affect the project's NPV and BCR.	Implement strict project management with contingency funds, experienced contractors, real-time tracking, and milestone-based payments to control delays and cost overruns.

Variability of water inflow	Assess how seasonal or long-term rainfall changes affect water diversion, hydropower output, and irrigation supply.	Use hydrological forecasting, adapt reservoir operations, add storage, and plan water releases based on climate projections to ensure year-round supply.
Environmental and social risks	Assess how biodiversity loss, groundwater use, and resettlement issues affect the project's acceptance and long-term sustainability.	Strengthen environmental monitoring, reforest and control slope erosion, and fairly restore displaced communities' livelihoods to secure social acceptance.

16. References

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